

Propuesta de Tesis

01 Información General

Título *Evaluation of graph distance functions for distribution comparison under perturbation*

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02 Problema y Motivación

Comparing distributions of graphs is a fundamental task in graph learning. It appears in graph generative model evaluation, distribution shift detection, and structural representation analysis, with applications in drug discovery, materials design, and synthetic data generation. The standard approach is Maximum Mean Discrepancy (MMD) with heuristic graph descriptors such as degree histograms, clustering coefficients, and orbit counts. However, O'Bray et al. (2022) showed that MMD rankings change depending on kernel and parameter choices. This makes comparative conclusions unreliable. Thompson et al. (2022) confirmed the problem and proposed alternatives, with mixed results. Several principled graph distance functions have been proposed as alternatives. These include Weisfeiler-Lehman kernels (Shervashidze et al., 2011), magnitude-based distances (Limbeck et al., 2024), and more recently Homomorphism Distortion (d_{HD}, Carrasco et al., 2025). Each captures different structural aspects and offers different tunability.

03 Objetivo e Hipótesis

Systematically evaluate principled graph distance functions for distribution comparison under controlled perturbations, comparing Homomorphism Distortion, magnitude-based distance, Weisfeiler-Lehman kernel distance, and MMD with standard descriptors. Characterize the sensitivity, monotonicity, and computational cost of each distance function across perturbation types, and study how tunable hyperparameters (pattern family F for d_{HD}, iteration depth h for WL) affect behavior.

The hypothesis is that d_{HD}, through its explicit parameterization by a pattern family F and its native coupling of feature and structural information, offers richer and more controllable sensitivity than alternative distance functions with simpler tunability.

04 Enfoque y Metodología

We follow the perturbation-sensitivity methodology of O'Bray et al. (2022) and Thompson et al. (2022). Given a graph distribution D and a perturbed version D' at varying intensities, we compute the distance between them using each candidate function and measure how that distance responds to perturbation. A good distance function should respond monotonically and sensitively. The distance functions compared are d_{HD} (Carrasco et al., 2025) with multiple pattern families F (paths, cycles, cycles plus small cliques), magnitude-based distance (Limbeck et al., 2024), Weisfeiler-Lehman kernel distance (Shervashidze et al., 2011) at multiple iteration depths, and MMD with standard descriptors as the current practice baseline. Distribution-level distances are computed via MMD-style aggregation using each candidate function. The perturbations are restricted to two types for depth: random edge rewiring via degree-preserving shuffling, chosen as a generic structural perturbation, and mode collapse simulation following Thompson et al. (2022). Experiments run on synthetic graph distributions (Erdos-Renyi, stochastic block models, Barabasi-Albert) for controlled ground truth, and small real datasets (ZINC, QM9 subsets) for practical validation. Each distance function is evaluated on sensitivity, monotonicity, and computational cost.

05 Resultados Esperados y Alcance

PFC1: Literature review on graph distance functions, perturbation-sensitivity methodology, and distribution comparison evaluation. Implementation of the evaluation pipeline with all four distance function families. Preliminary perturbation-sensitivity results on one perturbation type across synthetic graph distributions, with initial comparison across distance functions.

PFC2: Full perturbation-sensitivity analysis across both perturbation types and tunable hyperparameters (F for d_{HD}, h for WL). Characterization of which distance function is most sensitive to which perturbation type. Computational cost characterization. Empirical guidance for distance function selection given a perturbation type, framed honestly as dataset-specific findings with discussion of which aspects may generalize. Optional secondary validation using classifier-based aggregation (PolyGraph).

06 Referencias Clave

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07 Validación

	Nombre	Estado	Observaciones
R1	[Nombre]	Seleccionar ▾	[Observaciones]
R2	[Nombre]	Seleccionar ▾	[Observaciones]
R3	[Nombre]	Seleccionar ▾	[Observaciones]